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# Bivariate Birnbaum-Saunders accelerated lifetime model: estimation and diagnostic analysis

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## ABSTRACT

In this paper, we discuss the bivariate Birnbaum-Saunders accelerated lifetime model, in which we have modeled the dependence structure of bivariate survival data through the use of frailty models. Specifically, we propose the bivariate model Birnbaum-Saunders with the following frailty distributions: gamma, positive stable and logarithmic series. We present a study of inference and diagnostic analysis for the proposed model, more concisely, are proposed a diagnostic analysis based in local influence and residual analysis to assess the fit model, as well as, to detect influential observations. In this regard, we derived the normal curvatures of local influence under different perturbation schemes and we performed some simulation studies for assessing the potential of residuals to detect misspecification in the systematic component, the presence in the stochastic component of the model and to detect outliers. Finally, we apply the methodology studied to real data set from recurrence in times of infections of 38 kidney patients using a portable dialysis machine, we analyzed these data considering independence within the pairs and using the bivariate Birnbaum-Saunders accelerated lifetime model, so that we could make a comparison and verify the importance of modeling dependence within the times of infection associated with the same patient.

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## 1. Introduction

The Birnbaum-Saunders (BS) distribution is a lifetime distribution developed by Birnbaum and Saunders [2] to analyze material fatigue data. From its origin to the present day, this distribution has been studied under different aspects, besides gaining prominence through applications in other areas of knowledge (see, for example, [18]). According to [30], most of the works related to the BS models were developed assuming that the experimental units are independent. However, some works on correlated BS data has been developed in recent years, for example, Marchant et al. [20] developed multivariate logarithmic generalized BS distributions and multivariate generalized BS regression models for describing fatigue data. Multivariate generalized BS regression models and diagnostic measures were developed by Marchant et al. [21]. An alternative approach for analyzing multivariate (correlated)

BS data based on estimating equations was developed by Tsuyuguchi et al. [30]. Specifically, for the bivariate case, the bivariate BS distribution and some inferential aspects were introduced by Kundu et al. [14], and Vilca et al. [31] derived a robust extension of the bivariate BS distribution. Also, Saulo et al. [29] introduced a bivariate BS distribution based on the reparameterized BS distribution proposed by Santos-Neto et al. [28]. Kundu [15] developed the bivariate log BS distribution and different properties were obtained using copula structure. Vilca et al. [33] proposed a bivariate BS regression model through the use of bivariate sinh-normal (SN) distribution, and [32] defined the bivariate Sinh-Elliptical distribution and discussed some of its properties.

Frailty models have been receiving special attention in recent decades to model dependence. For instance, Hougaard [12] presented a general approach to frailty models for survival data, Giussani and Bonetti [8] investigated a new family of parametric bivariate frailty models, Oakes [24] discussed bivariate survival models induced by frailties, Liu et al. [19] proposed frailty proportional hazards models for the recurrent event processes where the dependence is modeled by conditioning on shared frailty included in both hazard functions, Sankaran and Anisha [27] presented a shared frailty model for gap time distributions of recurrent events with multiple causes, Sahu et al. [26] considered a Weibull regression model with gamma frailties for multivariate survival data, Leão et al. [16] proposed a methodology based on a BS frailty regression model that can be applied to censored or uncensored data and [17] proposed a survival model with frailty based on the BS distribution. Furthermore, Choi and Matthews [4] proposed a bivariate accelerated lifetime model, where the gamma, positive stable, power series, logarithmic series, and lognormal frailty distributions are considered. However, for the bivariate BS regression model, there is no study using the frailty approach given in [4].

This paper aims to propose a bivariate BS accelerated lifetime model (bivariate BSAL model) with a dependence structure modeled via the frailty approach and present diagnostic methods based on local influence and residual analysis for the proposed model.

This paper is organized as follows. In Section 2, we present the BS distribution and its relationship with the SN distribution. We also propose the bivariate BSAL model and discuss inferential procedures based on the maximum likelihood method for such a model. In Section 3, we derive some diagnostic procedures based on residuals and local influence. In Section 4, we perform some Monte Carlo simulation studies for assessing the potential of the residuals to detect misspecification. In Section 5, we present and explore an empirical application to show the applicability of the proposed model. Some concluding remarks are given in Section 6. Finally, some derivations are presented in the Appendix.

## 2. Modeling

In this section, we present a brief review about the BS distribution. We introduce the bivariate BSAL model as well as we also present estimation method and we discuss asymptotic inference aspects.

### 2.1. The BS distribution

Let  $T$  be a continuous nonnegative random variable, we say that  $T$  has BS distribution with parameters  $\alpha$  and  $\beta$ , which we will denote by  $T \sim \text{BS}(\alpha, \beta)$ , if its probability density

function (pdf) is given by:

$$f(t) = \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{1}{2\alpha^2} \left( \frac{t}{\beta} + \frac{\beta}{t} - 2 \right) \right\} \frac{t^{-\frac{3}{2}}(t + \beta)}{2\alpha\sqrt{\beta}}, \quad \forall t > 0.$$

In this case,  $\alpha > 0$  is the shape parameter and  $\beta > 0$  is the scale parameter. It can be shown that the survival function of the BS distribution is given by

$$S(t) = 1 - \Phi \left[ \frac{1}{\alpha} \left( \sqrt{\frac{t}{\beta}} - \sqrt{\frac{\beta}{t}} \right) \right],$$

where  $\Phi(\cdot)$  denotes the cumulative distribution function (cdf) of the standard normal distribution.

Hence, an important property establishes a relationship between the BS distribution and the SN distribution. Such property says that if  $T \sim \text{BS}(\alpha, \beta)$  then  $Y = \log(T) \sim \text{SN}(\alpha, \mu, 2)$ , whose pdf and cdf are given, respectively, by

$$f(y) = \frac{1}{\alpha} \cosh \left( \frac{y - \mu}{2} \right) \phi \left[ \frac{2}{\alpha} \sinh \left( \frac{y - \mu}{2} \right) \right]$$

and

$$F(y) = \Phi \left[ \frac{2}{\alpha} \sinh \left( \frac{y - \mu}{2} \right) \right],$$

where  $\phi(\cdot)$  pdf of the standard normal distribution and  $\mu = \log(\beta)$ , with  $\alpha, \sigma > 0$  and  $y \in \mathbb{R}$ .

Due to the relationship established between the above distributions, the distribution of  $Y = \log(T)$  is also called the log-Birnbaum-Saunders distribution (log-BS). For more details see [25].

## 2.2. Accelerated lifetime model

Assuming that the survival time of the  $i$ th subject ( $i = 1, \dots, n$ ) in the  $j$ th group ( $j = 1, 2$ ) is denoted by  $T_{ij}$ , with  $T_{ij} \sim \text{BS}(\alpha, \exp(\mathbf{x}_{ij}^\top \boldsymbol{\beta}))$ , where  $\mathbf{x}_{ij}^\top$  denotes the observations on  $k$  explanatory variables and  $\boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_k)^\top$  denotes the vector of regression coefficients, we consider log-linear regression model

$$Y_{ij} = \log(T_{ij}) = \mathbf{x}_{ij}^\top \boldsymbol{\beta} + W_{ij}, \quad (1)$$

where  $W_{ij} = \log(\epsilon_{ij}) \sim \log - \text{BS}(\alpha, 0, 2)$ . So, we will assume independence among the subjects and dependence within groups. To model this dependence, we will consider the frailty model, defined in [24], for  $(W_{i1}, W_{i2})^\top$  and thus, the bivariate survival function of the log-BS model stochastic components is given by

$$S(w_{i1}, w_{i2}) = \mathcal{L} \left\{ -\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})] \right\}, \quad (2)$$

where  $\xi_{2j} = (2/\alpha) \sinh(w_{ij}/2)$  and  $\mathcal{L}$  is the Laplace transform of the distribution of the pair-specific frailties, which depends on a frailty parameter,  $\kappa$ . In this paper, we will use

the following frailty distributions: gamma, positive stable and logarithmic series. Moreover, to measure dependence within groups we used the Kendall's tau, given by  $\tau = 1 - 4 \int_0^\infty \int_0^\infty \mathcal{L}'(g)^2 dg dv$ , where  $g = -\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]$  (see, [3]). The expressions for Kendall's tau and Laplace transforms of the frailty variable may be found in [3,11].

Let  $(y_{1j}, \delta_{1j}), \dots, (y_{nj}, \delta_{nj})$  be an observed sample of  $n$  independent observations, where  $y_{ij}$  is the logarithm of response measurement  $j$  in pair  $i$  and  $\delta_{ij}$  is the failure indicator variable, with  $i = 1, \dots, n$  and  $j = 1, 2$ . Then, the log-likelihood function is given by

$$l(\theta) = \sum_{i=1}^n \left[ \delta_{i1} \delta_{i2} \log(f(w_{i1}, w_{i2})) + \delta_{i1} (1 - \delta_{i2}) \log \left[ -\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}} \right] \right. \\ \left. + (1 - \delta_{i1}) \delta_{i2} \log \left[ -\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}} \right] + (1 - \delta_{i1}) (1 - \delta_{i2}) \log[S(w_{i1}, w_{i2})] \right],$$

where  $w_{ij} = y_{ij} - (\mathbf{x}_{ij}^\top \boldsymbol{\beta})$ ,  $\boldsymbol{\theta} = (\boldsymbol{\beta}, \kappa, \alpha)^\top$ ,  $f(w_{i1}, w_{i2}) = \xi_{11} \xi_{12} K(\xi_{21}) K(\xi_{22}) \mathcal{L}''\{-\log(1 - \Phi(\xi_{21})) - \log[1 - \Phi(\xi_{22})]\}$  and  $\partial/\partial w_{ij}(S(w_{i1}, w_{i2})) = \xi_{1j} K(\xi_{2j}) \mathcal{L}'\{-\log(1 - \Phi(\xi_{21})) - \log[1 - \Phi(\xi_{22})]\}$ , with  $\xi_{1j} = (2/\alpha) \cosh(w_{ij}/2)$  and  $K(\xi_{2j}) = \phi(\xi_{2j})/\{2[1 - \Phi(\xi_{2j})]\}$ .

The score functions for  $\boldsymbol{\beta}$ ,  $\kappa$  and  $\alpha$  are, respectively, given by

$$U_{\boldsymbol{\beta}}(\boldsymbol{\theta}) = -\left[ \mathbf{X}_1^\top \mathbf{D}(\mathbf{a}_1) + \mathbf{X}_2^\top \mathbf{D}(\mathbf{a}_2) \right] \mathbf{Z},$$

$$U_{\kappa}(\boldsymbol{\theta}) = \text{tr}(\mathbf{D}(\mathbf{b}))$$

and

$$U_{\alpha}(\boldsymbol{\theta}) = \text{tr}(\mathbf{D}(\mathbf{c})),$$

where  $\mathbf{X}_j$  is a matrix of dimension  $n \times (k+1)$ , with the  $i$ th line given by  $\mathbf{x}_{ij} = (x_{ij}^0, \dots, x_{ij}^k)^\top$ ,  $\mathbf{Z} = [1, \dots, 1]^\top$  and  $\mathbf{D}(\cdot)$  represents a diagonal matrix. The quantities  $\mathbf{a}_1(\cdot)$ ,  $\mathbf{a}_2(\cdot)$ ,  $\mathbf{b}(\cdot)$  and  $\mathbf{c}(\cdot)$  may be found in the Appendix.

The maximum likelihood estimates are solutions of the equation system  $U_{\boldsymbol{\beta}}(\boldsymbol{\theta}) = \mathbf{0}$ ,  $U_{\kappa}(\boldsymbol{\theta}) = 0$  and  $U_{\alpha}(\boldsymbol{\theta}) = 0$ . On the other hand, such equations do not present closed solutions and the use of iterative methods is necessary to find the roots. In this study, we used the BFGS method from the `maxLik` package of the software R. As initial values we have considered the least squares estimates for  $\boldsymbol{\beta}$ , for  $\kappa$  we obtained as a function of the initial value for the Kendall's tau and, finally, for  $\alpha$  we used the expression

$$\tilde{\alpha} = \left\{ \frac{4}{n} \sum_{i=1}^n \left[ \sinh^2 \left( \frac{y_{i1} - \mathbf{x}_{i1}^\top \boldsymbol{\beta}}{2} \right) + \sinh^2 \left( \frac{y_{i2} - \mathbf{x}_{i2}^\top \boldsymbol{\beta}}{2} \right) \right] \right\}^{\frac{1}{2}}. \quad (3)$$

So, the inference for the parameter vector  $\boldsymbol{\theta}$  can be based on the asymptotic distribution of the maximum likelihood estimator,  $\hat{\boldsymbol{\theta}}$ , that under suitable regularity conditions (see [6]), is given by

$$\hat{\boldsymbol{\theta}} \stackrel{a}{\sim} N_{k+3}(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{\theta}}), \quad (4)$$

where  $\boldsymbol{\Sigma}_{\boldsymbol{\theta}}$  denotes the asymptotic variance-covariance matrix for  $\hat{\boldsymbol{\theta}}$ , which may be approximated by the observed information matrix evaluated at  $\hat{\boldsymbol{\theta}}$ , given in the Appendix.

### 3. Diagnostic analysis

Indeed, an important step in the validation of a regression model is the verification of possible departures from the model assumptions, more specifically from the stochastic and systematic components of the model, as well as, to detect discrepant observations (outliers) that present some influence on the parameter estimates that may cause inferential changes. In this regard, Cook [5] presents the local influence method as a general way for assessing the joint influence of observations, under small perturbations in the model or data. Then, we will define some residuals for assessing the fit of the model proposed and we will make a study of local influence, for some perturbation schemes.

#### 3.1. Residual analysis

In order to verify the quality of the fit of the bivariate BSAL model, we will consider the  $V$ -residuals and  $K$ -residuals proposed by Choi and Matthews [4]. The  $V$ -residuals are defined by

$$\begin{aligned}\widehat{V}_i &= \mathcal{L} \{ -\log [S_1(\widehat{W}_{i1})] - \log [S_2(\widehat{W}_{i2})] \} \\ &= \mathcal{L} \{ \mathcal{L}^{-1}(\widehat{U}_{i1}) + \mathcal{L}^{-1}(\widehat{U}_{i2}) \},\end{aligned}\quad (5)$$

where  $\widehat{U}_{ij} = \mathcal{L}\{-\log[S_j(\widehat{W}_{ij})]\}$ , with  $S_j(\widehat{W}_{ij}) = 1 - (2/\alpha) \sinh(\widehat{W}_{ij}/2)$  and  $\widehat{W}_{ij} = y_{ij} - \mathbf{x}_{ij}^\top \widehat{\boldsymbol{\beta}}$ , for  $i = 1, \dots, n$  and  $j = 1, 2$ . Furthermore,  $\mathcal{L}^{-1}$  denotes the inverse of the Laplace transform of the considered frailty distribution.

Note that  $C(\widehat{u}_{i1}, \widehat{u}_{i2})$  have the form of an Archimedean copula with generator  $\varphi = \mathcal{L}^{-1}$ . Hence, choosing a frailty distribution that satisfies the conditions required for  $C(\widehat{u}_{i1}, \widehat{u}_{i2})$  to be an Archimedean copula, it follows from a result presented in [7] that the  $V$ -residuals have a distribution given by

$$G(v_i) = P(V_i < v_i) = v_i - \frac{\varphi(v_i)}{\varphi'(v_i)},$$

which depends on a frailty parameter  $\kappa$ .

Since the distribution of the  $V$ -residuals is not fully specified, as it depends on the frailty parameter  $\kappa$ , the authors suggest a new residual defined by an integral probability transformation of the  $V$ -residuals, i.e.  $\widehat{K}_i = G(\widehat{V}_i)$ . Thus, it follows from the Total Probability Theorem that  $K$ -residuals have uniform distribution in the interval  $[0, 1]$ .

In the presence of right censoring, we consider  $V$ -residuals and adapted  $K$ -residuals, as in [4], given by

$$\widehat{V}_i^{\text{adap}} = \mathcal{L}\{\mathcal{L}^{-1}(\widehat{U}_{i1}^{\text{adap}}) + (\widehat{U}_{i2}^{\text{adap}})\}$$

and

$$\widehat{K}_i^{\text{adap}} = G(\widehat{V}_i^{\text{adap}}), \quad (6)$$

with

$$\widehat{U}_{ij}^{\text{adap}} = \widehat{U}_{ij} + \frac{1}{2}(\delta_{ij} - 1)\widehat{U}_{ij}, \quad (7)$$

where  $\delta_{ij}$  is the failure indicator variable. More details may be found in [4].

### 3.2. Local influence

The local influence, proposed by Cook [5], corresponds to one of the most modern diagnostic methods. The main idea of this method is to perform small perturbations on data or model and analyze the behavior of a specific measure associated with those perturbations. For this, we consider the log-likelihood function of a given model,  $l(\boldsymbol{\theta})$ , where  $\boldsymbol{\theta}$  is a vector of dimension  $p$ . Moreover, we consider  $\boldsymbol{\lambda}$  a  $n \times 1$  perturbation vector restricted to some open subset  $\Omega \subset \mathbb{R}^n$ . Thereby, the log-likelihood function of the perturbed model is denoted by  $l(\boldsymbol{\theta} \mid \boldsymbol{\lambda})$ . Let  $\boldsymbol{\lambda}_0$  be a non-perturbation vector, we have  $l(\boldsymbol{\theta} \mid \boldsymbol{\lambda}_0) = l(\boldsymbol{\theta})$ . Hence, the likelihood displacement  $LD(\boldsymbol{\lambda}) = 2[l(\hat{\boldsymbol{\theta}}) - l(\hat{\boldsymbol{\theta}} \mid \boldsymbol{\lambda})]$  is used as a measure of influence, where  $\hat{\boldsymbol{\theta}} \mid \boldsymbol{\lambda}$  denotes the maximum likelihood estimate under the perturbed model. Further, using the concepts of Differential Geometry, Cook [5] shows that the normal curvature in the unit direction  $\mathbf{d}$  is given by  $C_d(\boldsymbol{\theta}) = 2|\mathbf{d}^\top \boldsymbol{\Delta}^\top \ddot{\mathbf{L}}_{\boldsymbol{\theta}\boldsymbol{\theta}}^{-1} \boldsymbol{\Delta} \mathbf{d}|$ , where  $\ddot{\mathbf{L}}_{\boldsymbol{\theta}\boldsymbol{\theta}}^{-1}$  is the observed information matrix associated with the considered model and  $\boldsymbol{\Delta}$  is a  $p \times n$  matrix that depends on the perturbation scheme with elements  $\Delta_{rs} = \partial^2 l(\boldsymbol{\theta} \mid \boldsymbol{\lambda}) / \partial \theta_r \partial \lambda_s$ , with  $r = 1, \dots, p$  and  $s = 1, \dots, n$ , evaluated at  $\hat{\boldsymbol{\theta}}$  and  $\boldsymbol{\lambda}_0$ . So, Cook [5] suggests using the  $\mathbf{d}_{\max}$  direction, which corresponds to the largest curvature,  $C_{d_{\max}}(\boldsymbol{\theta})$ .

Some diagnostic plots are suggested, for example, the plot of  $C_{d_r}(\boldsymbol{\theta})$  against the order of observations, where  $\mathbf{d}_r$  is a vector  $n \times 1$  composed of one in the  $r$ th row and zero in the others. Such curvature is denoted by  $C_r(\boldsymbol{\theta})$  and is given by  $C_r(\boldsymbol{\theta}) = 2|\boldsymbol{\Delta}_r^\top \ddot{\mathbf{L}}_{\boldsymbol{\theta}\boldsymbol{\theta}}^{-1} \boldsymbol{\Delta}_r|$ , where  $\boldsymbol{\Delta}_r^\top$  is the  $r$ th row of  $\boldsymbol{\Delta}$ . Besides, a careful note observation in the observations is suggested such that  $C_r(\boldsymbol{\theta}) > 2\bar{C}$ , where  $\bar{C} = (1/n) \sum_{r=1}^n C_r(\boldsymbol{\theta})$ .

As follows, we will calculate for the perturbation schemes considered, the expressions for the matrix components

$$\boldsymbol{\Delta} = \left( \frac{\partial^2 l(\boldsymbol{\theta} \mid \boldsymbol{\lambda})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\lambda}^\top} \right),$$

considering the model defined in (1) and (2) and its log-likelihood function given by (3).

*Case-weights perturbation:* Let  $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_n)^\top$  be the vector of perturbation. In this context, perturbed log-likelihood function is given by

$$l(\boldsymbol{\theta} \mid \boldsymbol{\lambda}) = \sum_{i=1}^n \lambda_i l_i(\mathbf{w}_i^\top; \boldsymbol{\theta}),$$

with  $0 \leq \lambda_i \leq 1$ ,  $\boldsymbol{\lambda}_0 = (1, \dots, 1)^\top$  and

$$\begin{aligned} l_i(\mathbf{w}_i^\top; \boldsymbol{\theta}) &= \delta_{i1} \delta_{i2} \log(f(w_{i1}, w_{i2})) + \delta_{i1} (1 - \delta_{i2}) \log \left[ -\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}} \right] + \delta_{i2} \\ &\quad \times (1 - \delta_{i1}) \log \left[ -\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}} \right] + (1 - \delta_{i1})(1 - \delta_{i2}) \log [S(w_{i1}, w_{i2})], \end{aligned}$$

where  $\mathbf{w}_i = (w_{i1}, w_{i2})^\top$ . In this case  $\boldsymbol{\Delta} = (\boldsymbol{\Delta}_\beta, \boldsymbol{\Delta}_\kappa, \boldsymbol{\Delta}_\alpha)^\top$ , with  $\boldsymbol{\Delta}_\beta = -[X_1^\top \mathbf{D}(\mathbf{a}_1) + X_2^\top \mathbf{D}(\mathbf{a}_2)]$ ,  $\boldsymbol{\Delta}_\kappa = (b_1(\mathbf{w}_1^\top, \boldsymbol{\theta}), \dots, b_n(\mathbf{w}_n^\top, \boldsymbol{\theta}))^\top$  and  $\boldsymbol{\Delta}_\alpha = (c_1(\mathbf{w}_1^\top, \boldsymbol{\theta}), \dots, c_n(\mathbf{w}_n^\top, \boldsymbol{\theta}))^\top$ , where  $X_j$ ,  $\mathbf{D}(\cdot)$ ,  $a_{ij}(\cdot)$ ,  $b_i(\cdot)$  and  $c_i(\cdot)$  are given in the Appendix, for  $i = 1, \dots, n$  and  $j = 1, 2$ , with  $a_{ij}(\cdot)$ ,  $b_i(\cdot)$  and  $c_i(\cdot)$  evaluated at  $\hat{\boldsymbol{\theta}}$  and  $\boldsymbol{\lambda}_0$ .

*Perturbation of the response variable:* Consider that each  $w_{i1}$  and  $w_{i2}$  is perturbed as  $w_{i1}(\lambda) = w_{i1} + \lambda_i$  and  $w_{i2}(\lambda) = w_{i2} + \lambda_i$ , respectively, where  $w_{ij} = y_{ij} - (\mathbf{x}_{ij}^\top \boldsymbol{\beta})$ , with  $i = 1, \dots, n$  and  $j = 1, 2$ . Thus, perturbed log-likelihood function is given by

$$l(\boldsymbol{\theta} | \boldsymbol{\lambda}) = \sum_{i=1}^n l_i(\mathbf{w}_i(\boldsymbol{\lambda})^\top; \boldsymbol{\theta}),$$

with  $\mathbf{w}_i(\boldsymbol{\lambda}) = (w_{i1}(\boldsymbol{\lambda}), w_{i2}(\boldsymbol{\lambda}))^\top$  and  $\boldsymbol{\lambda}_0 = (0, \dots, 0)^\top$  is the non-perturbation vector. Hence, the components of matrix  $\boldsymbol{\Delta}$  are  $\boldsymbol{\Delta}_\beta = -[\mathbf{X}_1^\top \mathbf{D}(\mathbf{k}_1) + \mathbf{X}_2^\top \mathbf{D}(\mathbf{k}_2)]$ ,  $\boldsymbol{\Delta}_\kappa = (l_1, \dots, l_n)^\top$  and  $\boldsymbol{\Delta}_\alpha = (m_1, \dots, m_n)^\top$ , with  $\mathbf{D}(\mathbf{k}_j) = \text{diag}\{k_{1j}, \dots, k_{nj}\}$ , for  $k_{ij} = \partial/\partial\lambda_i[a_{ij}(\mathbf{w}_i(\boldsymbol{\lambda})^\top; \boldsymbol{\theta})]$ ,  $l_i = \partial/\partial\lambda_i[b_i(\mathbf{w}_i(\boldsymbol{\lambda})^\top; \boldsymbol{\theta})]$  and  $m_i = \partial/\partial\lambda_i[c_i(\mathbf{w}_i(\boldsymbol{\lambda})^\top; \boldsymbol{\theta})]$ , where  $a_{ij}(\cdot)$ ,  $b_i(\cdot)$  and  $c_i(\cdot)$  are given in the Appendix and evaluated at  $\hat{\boldsymbol{\theta}}$  and  $\boldsymbol{\lambda}_0$ .

We can perturb the response variable only in one coordinate. To do so, we must consider  $\mathbf{w}_i(\boldsymbol{\lambda}) = (w_{i1} + \lambda_i; w_{i2})^\top$  or  $\mathbf{w}_i(\boldsymbol{\lambda}) = (w_{i1}; w_{i2} + \lambda_i)^\top$ .

*Explanatory variable perturbation:* Consider now an additive perturbation on a particular continuous explanatory variable, namely  $x_{ij}^s$ , by making  $x_{i1}^s(\lambda) = x_{i1}^s + \lambda_i$  and  $x_{i2}^s(\lambda) = x_{i2}^s + \lambda_i$ , where  $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_n)^\top$ . In this regard,  $\boldsymbol{\lambda}_0 = (0, \dots, 0)^\top$  and  $w_{ij}(\boldsymbol{\lambda}) = y_{ij} - (\mathbf{x}_{ij}^\top(\boldsymbol{\lambda})\boldsymbol{\beta})$ , with  $\mathbf{x}_{ij}^\top(\boldsymbol{\lambda}) = (x_{ij}^0, \dots, (x_{ij}^s + \lambda_i), \dots, x_{ij}^k)$ . Then, the perturbed log-likelihood function takes here the form

$$l(\boldsymbol{\theta} | \boldsymbol{\lambda}) = \sum_{i=1}^n l_i(\mathbf{w}_i(\boldsymbol{\lambda})^\top; \boldsymbol{\theta}),$$

where  $\mathbf{w}_i(\boldsymbol{\lambda}) = (w_{i1}(\boldsymbol{\lambda}), w_{i2}(\boldsymbol{\lambda}))^\top$ , with  $i = 1, \dots, n$  and  $j = 1, 2$ . Thereby, the components of matrix  $\boldsymbol{\Delta}$  are  $\boldsymbol{\Delta}_\beta = -[\mathbf{X}_1^\top \mathbf{D}(\mathbf{k}'_1) + \mathbf{X}_2^\top \mathbf{D}(\mathbf{k}'_2)]$ ,  $\boldsymbol{\Delta}_\kappa = (l'_1, \dots, l'_n)^\top$  and  $\boldsymbol{\Delta}_\alpha = (m'_1, \dots, m'_n)^\top$ , with  $\mathbf{D}(\mathbf{k}'_j) = \text{diag}\{k'_{1j}, \dots, k'_{nj}\}$ , for  $k'_{ij} = \partial/\partial\lambda_i[a_{ij}(\mathbf{w}_i(\boldsymbol{\lambda})^\top; \boldsymbol{\theta})]$ ,  $l'_i = \partial/\partial\lambda_i[b_i(\mathbf{w}_i(\boldsymbol{\lambda})^\top; \boldsymbol{\theta})]$  and  $m'_i = \partial/\partial\lambda_i[c_i(\mathbf{w}_i(\boldsymbol{\lambda})^\top; \boldsymbol{\theta})]$ , where  $a_{ij}(\cdot)$ ,  $b_i(\cdot)$  and  $c_i(\cdot)$  are given in the Appendix and evaluated at  $\hat{\boldsymbol{\theta}}$  and  $\boldsymbol{\lambda}_0$ .

## 4. Numerical study

In this section, we performed some simulation studies for assessing the potential of  $K$ -residuals to detect misspecification in the systematic component, the presence in the stochastic component of the model, as well as, to detect outliers. The results were obtained through a computational routine implemented in R statistical software. Codes can be obtained from the authors upon request.

### 4.1. The algorithm

Firstly, we generate a bivariate (correlated) sample,  $(Y_{i1}, Y_{i2})^\top$ , where  $Y_{ij} \sim \log\text{-BS}(\alpha, \mathbf{x}_{ij}^\top \boldsymbol{\beta}, 2)$ , with  $i = 1, \dots, n$  and  $j = 1, 2$ , with based on the algorithm presented in [23]. As a result, we have the following algorithm:

To investigate the capacity of the  $K$ -residuals to highlight specific problems of model goodness-of-fit, we will perform a simulation study similar to [4] using the following



**Algorithm 1** Generation of a correlated bivariate sample  $(Y_{i1}, Y_{i2})^\top$ 

- 
- [1.] Generate two independent uniform  $(0, 1)$  variates  $t$  and  $u$ .
  - [2.] Obtain  $C_u(t)$ , so that  $C_u(t) = P(T \leq t | U = u) = \frac{\partial C(u, t)}{\partial u}$ , where  $C$  is the copula associated with  $W_{i1}$  and  $W_{i2}$  for the frailty distributions gamma (G), positive stable (PS) and logarithmic series (LS).
  - [3.] Define  $v = C_u^{-1}(t)$ , where  $C_u^{-1}$  denotes the inverse of  $C_u(t)$ .
  - [4.] Consider  $w_{i1} = S_j^{-1}(u)$  and  $w_{i2} = S_j^{-1}(v)$ ,  $i = 1, \dots, n$ , where  $S^{-1}$  denotes the inverse of the survival function of the log-BS( $\alpha, 0, 2$ ) distribution.
  - [5.] Generate  $\mathbf{x}_{ij}^\top$  from a standard normal distribution.
  - [6.] Obtain  $y_{ij}$ , so that  $y_{ij} = \mathbf{x}_{ij}^\top \boldsymbol{\beta} + w_{ij}$ , for  $i = 1, \dots, n$  and  $j = 1, 2$ .
- 

summary measure of fit:

$$S = \frac{1}{n} \sum_{i=1}^n \left\{ \widehat{K}_{(i)} - \frac{i}{n+1} \right\}^2, \quad i = 1, \dots, n. \quad (8)$$

In the presence of right censoring, we consider the adapted summary measure of fit, given by:

$$S_{\text{adap}} = \frac{1}{n} \sum_{i=1}^n \left\{ \widehat{K}_{(i)}^{\text{adap}} - \frac{i}{n+1} \right\}^2, \quad i = 1, \dots, n. \quad (9)$$

#### 4.2. A misspecified systematic component

Firstly, we consider a misspecified systematic component. In this case, we investigated if the presented residuals were able to detect the omission of important explanatory variables in the systematic part of the model. In order to do so, we consider the model:

$$Y_{ij} = \beta_0 + \sum_{s=1}^3 \beta_s x_{ij}^{(s)} + W_{ij}, \quad (10)$$

where  $W_{ij} \sim \log - \text{BS}(\alpha, 0, 2)$  and  $x_{ij}^{(s)} \sim N(0, 1)$ , with  $i = 1, \dots, n$  and  $j = 1, 2$ .

The sample sizes and true values of the parameters considered are  $n = 50, 100, 200$  and  $\beta_0 = 0, \beta_1 = 3, \beta_2 = 1.5, \beta_3 = -2$  and  $\alpha = 0.5$ . The values assigned to the frailty parameter  $\kappa$  are determined in function to the values assigned to the Kendall's tau, which in this case we consider 0.1, 0.3, 0.5 and 0.7. The number of Monte Carlo replications used was 5000, for each value assigned to the Kendall's tau. In addition, we use the gamma frailty distribution. Subsequently, we use the generated samples and fit the model to the data through four distinct scenarios:

- *Scenario 1:* We fitted the model (10), i.e. the model with all covariates.

**Table 1.** Measures  $S_T$  and  $S_W$  according to the scenarios considered.

| $\tau$ | Scenario 1 | Scenario 2 | Scenario3 | Scenario 4 |
|--------|------------|------------|-----------|------------|
| 0.1    | 0.000591   | 0.002411   | 0.004547  | 0.028376   |
| 0.3    | 0.000734   | 0.002160   | 0.007190  | 0.024706   |
| 0.5    | 0.001128   | 0.001993   | 0.003238  | 0.028080   |
| 0.7    | 0.001870   | 0.004271   | 0.006253  | 0.009526   |

- *Scenario 2:* We fitted the model  $Y_{ij} = \beta_0 + \sum_{s=1}^2 \beta_s x_{ij}^{(s)} + W_{ij}$ , i.e. we omitted the covariate  $x_{ij}^{(3)}$ .
- *Scenario 3:* We fitted the model  $Y_{ij} = \beta_0 + \beta_1 x_{ij}^{(1)} + W_{ij}$ , i.e. we omitted the covariates  $x_{ij}^{(2)}$  and  $x_{ij}^{(3)}$ .
- *Scenario 4:* We omitted all covariates from the model.

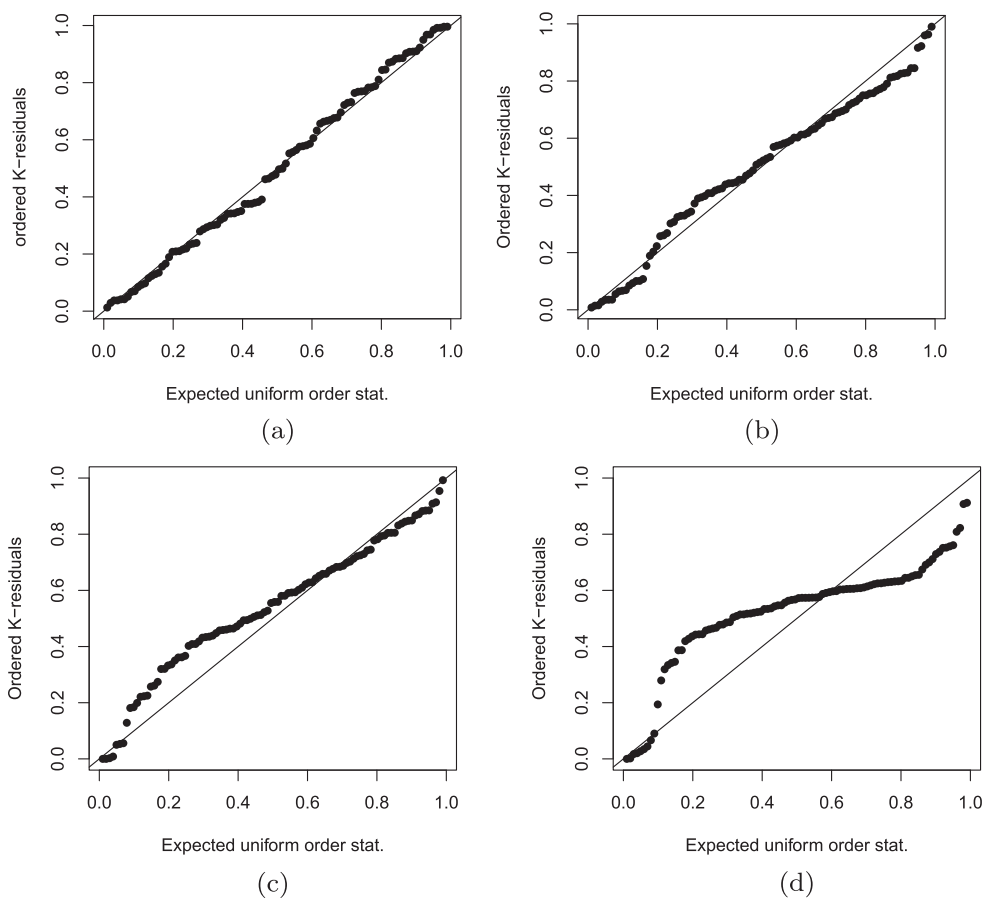
Then, we obtained the  $S_T$  and  $S_W$  measurements, which correspond to the fitted model with all covariates and with omitting covariates, respectively. In Table 1, we present the mean values of the measurements mentioned above for a sample size  $n = 100$  (results for samples size  $n = 50$  and  $n = 200$  were omitted for presenting similar results) in each scenario considered.

Based on Table 1, we can see that, in general, for fixed Kendall's tau values, the mean values of the measures considered increase according to we omitted more covariates from the model. Moreover, we can see that the average values of  $S_T$  increasing when we increase Kendall's tau values. Hence, we can conclude that the simulation study showed that the  $K$ -residuals were able to evidence the misspecification in the systematic component of the model.

In Figure 1 we present the Q-Q plots of the ordered  $K$ -residuals for a sample size  $n = 100$  and  $\tau = 0.3$  in each scenario considered. We can see that the Q-Q plot for Scenario 1 is very near to the reference diagonal, this means that the assumption where  $K$ -residuals have uniform distribution in the range  $[0, 1]$  is verified. Along with, we can also note that Q-Q plots gradually move away from the reference diagonal as we omitted more model explanatory variables. Therefore, the residuals are sensitive to misspecification in the systematic component of the model.

### 4.3. Presence of outlier

Likewise, we will also investigate whether  $K$ -residuals can detect the existence of outliers that may generate some influence on the parameter estimates that may cause inferential changes. To do so, we generated 5000 bivariate samples  $(Y_{i1}, Y_{i2})^\top$ ,  $i = 1, \dots, n$ , with sizes  $n = 50, 100$  and  $200$ , in which we used gamma frailty and considered as true parameters  $\beta_0 = 1.5$ ,  $\beta_1 = 3$  and  $\alpha = 0.5$ . Again, we assigned the values 0.1, 0.3, 0.5 and 0.7 to Kendall's tau and thus obtained the true values for the parameter  $\kappa$ . To sample contamination, consider  $(y_{i1}, y_{i2})^\top = ((7/5)y_{\arg\max_i(y_{i1})}, y_{i2})^\top$ , where  $\arg\max_i(y_{i1})$  returns the  $i$ th value such that  $y_{i1}$  is maximum, with  $i = 1, \dots, n$ . Thereby, we obtained the respective measurements,  $S_T$  and  $S_W$ , corresponding to the model adjustments to the case in which we did



**Figure 1.** Q-Q plots of ordered  $K$ -residuals considering misspecification in the systematic component of the model: (a) scenario 1, (b) scenario 2, (c) scenario 3 and (d) scenario 4.

**Table 2.** Measures  $S_T$  and  $S_W$  – Presence of Outlier.

| $\tau$ | $n = 50$ |          | $n = 100$ |          | $n = 200$ |          |
|--------|----------|----------|-----------|----------|-----------|----------|
|        | $S_T$    | $S_W$    | $S_T$     | $S_W$    | $S_T$     | $S_W$    |
| 0.1    | 0.001246 | 0.009017 | 0.000613  | 0.007489 | 0.000302  | 0.026853 |
| 0.3    | 0.001320 | 0.008143 | 0.000955  | 0.007715 | 0.000437  | 0.032681 |
| 0.5    | 0.001699 | 0.012168 | 0.001123  | 0.016090 | 0.000852  | 0.014765 |
| 0.7    | 0.002579 | 0.009703 | 0.001917  | 0.018154 | 0.001660  | 0.019405 |

not contaminate the sample and to the case in which there was contamination. We present the mean values of  $S_T$  and  $S_W$  in Table 2.

Through Table 2 we can see that the average values of measure  $S_W$  exceed the average values of measure  $S_T$  for all sample sizes considered, i.e. the  $K$ -residuals detect the presence of outlier.

4.4. A misspecified stochastic component

Now, we consider misspecification in the stochastic component of the model. To produce this misspecification, we generate the data by considering a frailty distribution, called true frailty, and adjust it by considering others frailty distributions. The values assigned to the parameters were  $\beta_0 = 0$ ,  $\beta_1 = 3$  and  $\alpha = 0.5$ . Again we assign the values 0.1, 0.3, 0.5 and 0.7 to Kendall's tau.

We generated 5000 samples of sizes 50, 100 and 200. Subsequently, we obtained the  $S_T$  and  $S_W$  measurements which correspond to the fitted model through the use of the true frailty and to the fitted model with distinct frailty distributions from the frailty distribution true, respectively. Table 3 shows the mean values of the  $S_T$  and  $S_W$  measurements for a sample size 100 (we omitted the results for sizes 50 and 100, as they were analogous).

As we can notice by Table 3, in general, the measures values remained near, regardless of the true frailty distribution considered. Thus, the  $K$ -residuals was not able to capture the misspecification in the stochastic component of the model.

4.5. Presence of right censoring

Now, we will study the behavior of adapted  $K$ -residuals and to achieve this we will make a comparison between the  $K$ -residuals and the adapted  $K$ -residuals in the presence of right censoring.

In this case, we use the same scenario considered in the misspecification of the stochastic component of the model. And to introduce it to the right censoring of the sample, we simulate these values of the lognormal distribution with variance 1 and with means 1.5, 1.0 and  $-0.5$  obtain 10%, 20% and 40% censoring.

We did this for one of the response variables of the generated bivariate sample and considered the other response variable of the pair as uncensored. Later, we adjusted the model and obtained the respective  $S$  and  $S_{\text{adap}}$  measures given in (8) and (9). Then, in Table 4 we present the averages of the  $S$  and  $S_{\text{adap}}$  measures for a sample size  $n = 100$ .

Hence, based on Table 4 we can see that, in general, when we fix the frailty distribution, the average values of the  $S$  and  $S_{\text{adap}}$  measures increase as we increase the percentage of censoring. Furthermore, we also noticed that the values attributed to Kendall's tau did not

Table 3. Measures  $S_T$  and  $S_W$  – misspecification on stochastic component.

| True frailty       | $\tau$ | Fitted frailty model |                 |                    |
|--------------------|--------|----------------------|-----------------|--------------------|
|                    |        | Gamma                | Positive stable | Logarithmic series |
| Gamma              | 0.1    | 0.000590             | 0.000670        | 0.000601           |
|                    | 0.3    | 0.000721             | 0.000807        | 0.000744           |
|                    | 0.5    | 0.001126             | 0.000968        | 0.000870           |
|                    | 0.7    | 0.001871             | 0.001404        | 0.001875           |
| Positive stable    | 0.1    | 0.000724             | 0.000608        | 0.000667           |
|                    | 0.3    | 0.001341             | 0.000634        | 0.001576           |
|                    | 0.5    | 0.002213             | 0.000764        | 0.005018           |
|                    | 0.7    | 0.002773             | 0.001588        | 0.016349           |
| Logarithmic series | 0.1    | 0.001184             | 0.000672        | 0.000766           |
|                    | 0.3    | 0.001270             | 0.000682        | 0.000819           |
|                    | 0.5    | 0.001428             | 0.000699        | 0.000926           |
|                    | 0.7    | 0.001745             | 0.000762        | 0.001231           |

**Table 4.** Measures  $S$  e  $S_{\text{adap}}$  – Presence of right censoring.

| Frailty | $\tau$ | 10%      |                   | 20%      |                   | 40%      |                   |
|---------|--------|----------|-------------------|----------|-------------------|----------|-------------------|
|         |        | $S$      | $S_{\text{adap}}$ | $S$      | $S_{\text{adap}}$ | $S$      | $S_{\text{adap}}$ |
| G       | 0.1    | 0.001218 | 0.000585          | 0.003123 | 0.000732          | 0.016263 | 0.007855          |
|         | 0.3    | 0.003177 | 0.001565          | 0.003313 | 0.000666          | 0.020950 | 0.005465          |
|         | 0.5    | 0.001651 | 0.000802          | 0.010454 | 0.003744          | 0.017202 | 0.005806          |
|         | 0.7    | 0.002048 | 0.001370          | 0.011765 | 0.005711          | 0.020369 | 0.003375          |
| PS      | 0.1    | 0.001156 | 0.000616          | 0.004232 | 0.002456          | 0.020873 | 0.017740          |
|         | 0.3    | 0.002175 | 0.001839          | 0.005035 | 0.002246          | 0.014630 | 0.010150          |
|         | 0.5    | 0.004207 | 0.003529          | 0.010284 | 0.009289          | 0.020428 | 0.012375          |
|         | 0.7    | 0.001591 | 0.001227          | 0.003620 | 0.002616          | 0.014710 | 0.004962          |
| LS      | 0.1    | 0.000844 | 0.000654          | 0.002086 | 0.000791          | 0.006873 | 0.001318          |
|         | 0.3    | 0.000991 | 0.000734          | 0.061754 | 0.044018          | 0.010035 | 0.004237          |
|         | 0.5    | 0.000827 | 0.000824          | 0.001572 | 0.000892          | 0.005856 | 0.001788          |
|         | 0.7    | 0.000861 | 0.001067          | 0.008665 | 0.004325          | 0.023288 | 0.011020          |

affect the magnitude of the measures considered. Finally, we found that for all censoring percentages considered, the average values of the  $S_{\text{adap}}$  measure were lower than the corresponding average values of the  $S$  measure. Thus, we conclude that the adapted  $K$ -residual is better than the  $K$ -residual for working with data involving the presence of right censoring.

## 5. Application

In this section, we consider a real data set corresponding to the recurrence times of infections of kidney patients using a portable dialysis machine. Initially, we present a brief description of the data. Next, we conduct an exploratory analysis, which suggests a positive asymmetric distribution to model the times of infection. Based on this, we compare independent BS and bivariate BSAL models and, finally, we perform local influence study for fitted bivariate BSAL model.

### 5.1. Description of the data

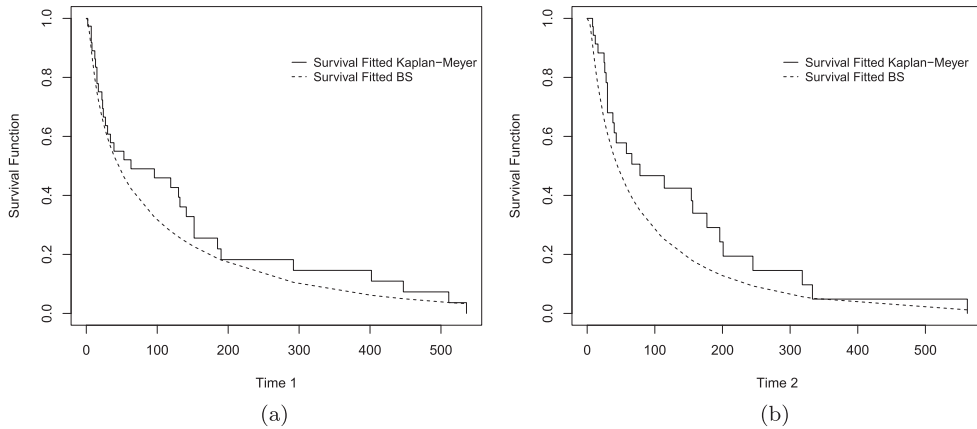
We consider the data set given in [13] which are related to the recurrence times of infections of 38 kidney patients using a portable dialysis machine. The data considered here has been analyzed by means of Cox proportional hazard model with a multiplicative frailty parameter for each patient [22], by means of a scale-location model for bivariate survival times, using the FGM copula [1], through of the Weibull/gamma shared frailty models [9] and through of the shared frailty models: inverse Gaussian, compound Poisson and compound negative binomial [10]. The aim of this study is to estimate the times of the first and second recurrence of renal infection for each individual, using the bivariate BSAL model, as given in (1) and (2), considering as explanatory variable the variable sex (0 = male, 1 = female). And the percentages of censored observations for the first and second infection time were 16% and 32%, respectively.

### 5.2. Exploratory data analysis

Table 5 provides a descriptive summary of the data considered that include standard deviation (SD), coefficients of variation (CV), skewness (CS) and kurtosis (CK). Figure 2 shows

**Table 5.** Descriptive statistics for the data.

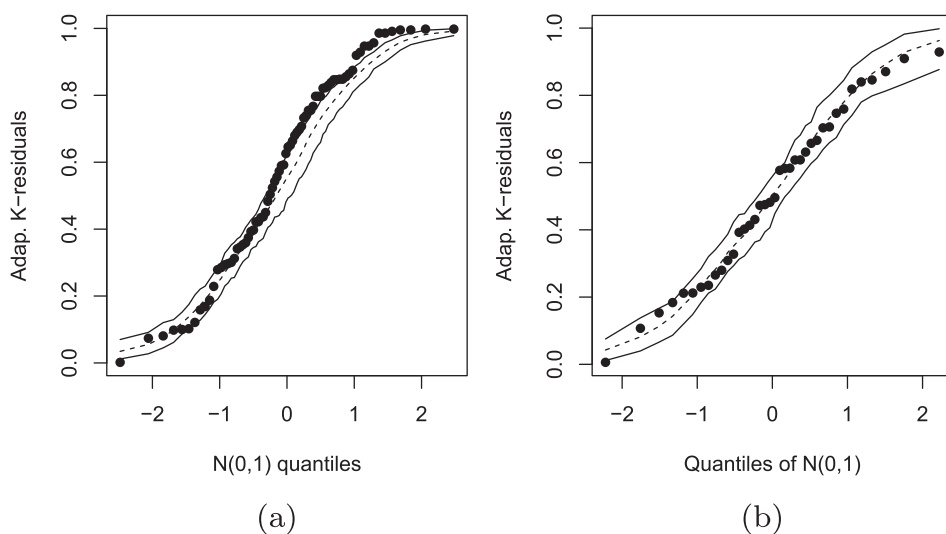
| Variable | Median | Mean   | SD      | CV      | CS   | CK   | Min  | Max    |
|----------|--------|--------|---------|---------|------|------|------|--------|
| Time 1   | 46.00  | 111.70 | 1444.01 | 128.91% | 1.69 | 1.92 | 2.00 | 536.00 |
| Time 2   | 39.00  | 91.55  | 117.44  | 128.28% | 2.10 | 4.82 | 4.00 | 562.00 |

**Figure 2.** Marginal survival estimates by Kaplan–Meier and BS distribution: (a) Time 1 and (b) Time 2.

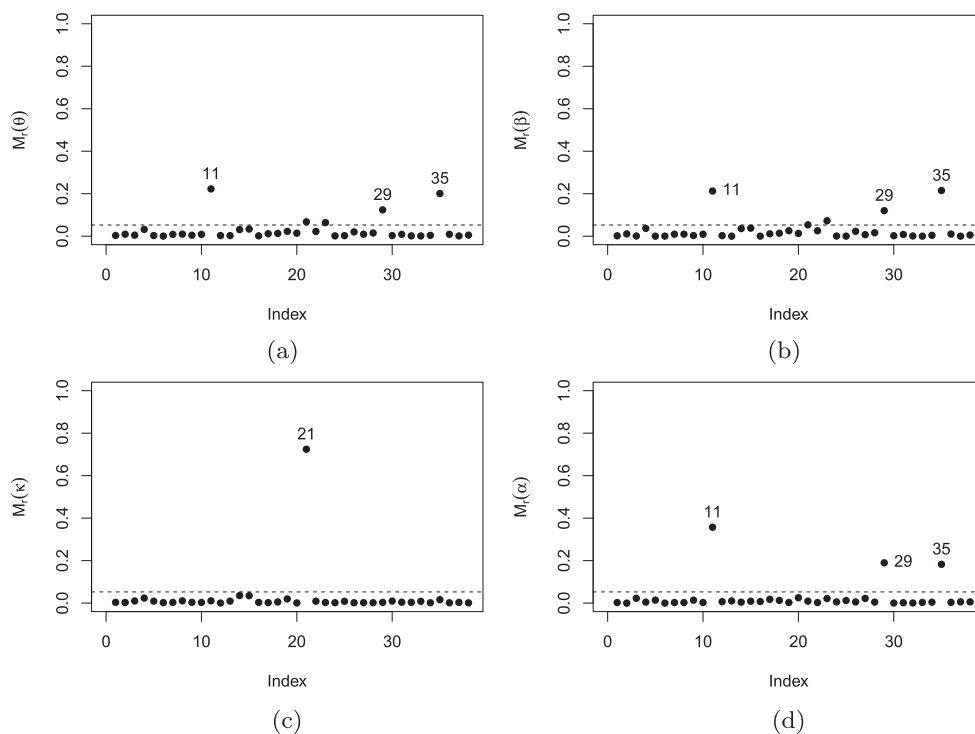
plots of the marginal fit of the events with BS distribution and also the non-parametric estimates are displayed. Through Table 5, observing the values of CS and CK, it is reasonable to admit that the times have a positively skewed distribution. Therefore, based on these observations and by Figure 2, we notice a BS distribution could be suitable for modeling marginal the data considered.

**Table 6.** Maximum likelihood estimates of the parameters (standard errors in parentheses and  $p$ -values in [·]) and the  $S_{adap}$ , AIC and BIC measures.

| Parameter  | Model       |                |
|------------|-------------|----------------|
|            | Independent | Bivariate BSAL |
| $\beta_0$  | 2.3242      | 1.8447         |
|            | (0.0573)    | (0.1012)       |
|            | [ < 0.05]   | [ < 0.05]      |
| $\beta_1$  | 1.0906      | 1.2833         |
|            | (0.0333)    | (0.0540)       |
|            | [ < 0.05]   | [ < 0.05]      |
| $\kappa$   | —           | 5.6126         |
|            | —           | (0.8267)       |
|            | —           | [0.093]        |
| $\alpha$   | 1.3967      | 1.2389         |
|            | (0.0147)    | (0.0252)       |
|            | —           | —              |
| $S_{adap}$ | 0.1501      | 0.1245         |
| AIC        | 204.0576    | 202.3188       |
| BIC        | 211.0498    | 207.2316       |



**Figure 3.** Normal probability plots with envelope – considering independence within pairs (a) and considering the bivariate BSAL model (b).



**Figure 4.** Index plots of  $M_r(\cdot)$  for: (a)  $\theta$ , (b)  $\beta$ , (c)  $\kappa$  and (d)  $\alpha$ , under the case-weights perturbation.

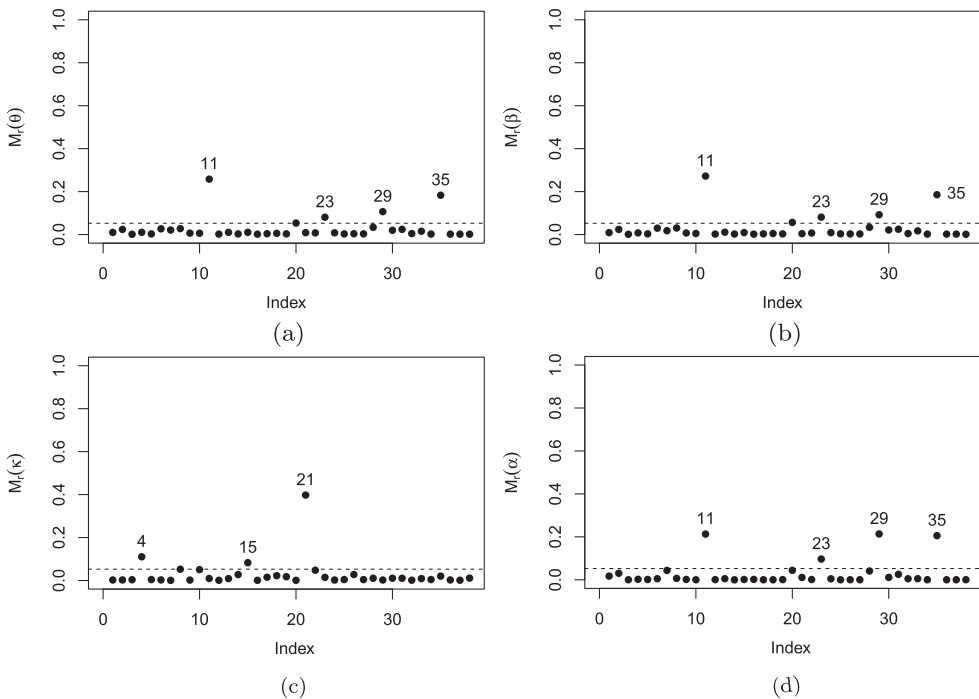
### 5.3. Modeling

Next, we will compare independent BS and BSAL models. We present in Table 6 the maximum likelihood estimates (standard errors in parentheses and  $p$ -values in  $[\cdot]$ ) of the model parameters considering independence within of the pairs observed and of the bivariate BSAL model and the statistics:  $S_{\text{adap}}$ , AIC and BIC. We observed that all parameters are significant at 10% for both fitted models and that all selection criteria indicate that the bivariate BSAL model is more suitable for analyzing these data.

To detect possible departures from the initial assumptions of the models, we present in Figure 3 the normal probability plots for the  $K$ -residuals with simulated envelope (for the independent case and for the bivariate BSAL model, respectively). Figure 3(a) shows an unusual behavior with points outside of the envelope. While Figure 3(b) shows an usual behavior that does not indicate serious deviation from assumption model.

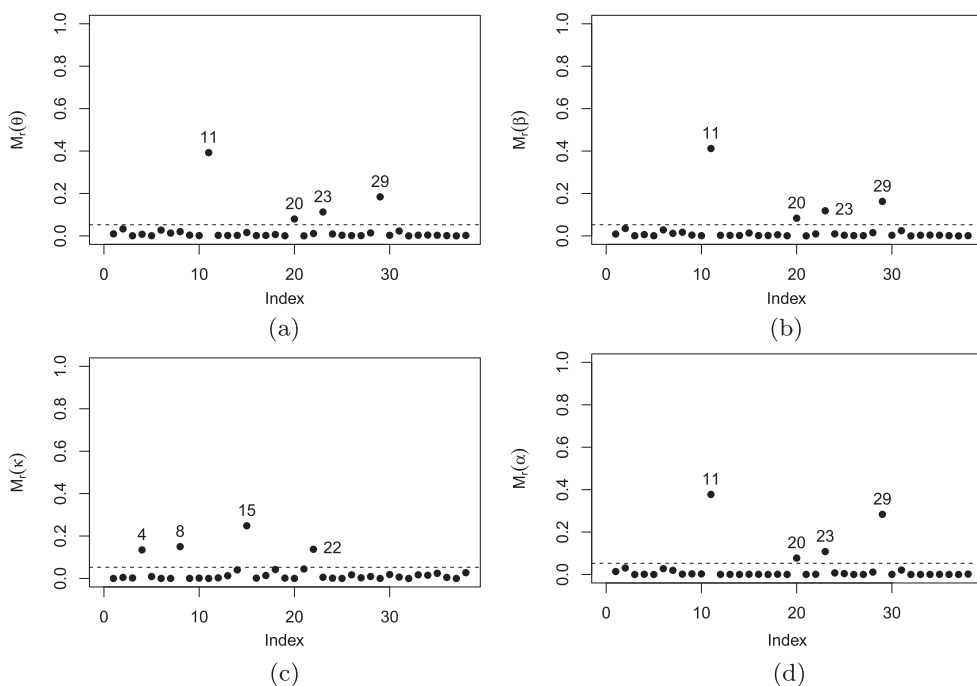
We performed local influence study for fitted bivariate BSAL model.

In Figures 4–7 we present the index plots of  $M_r(\theta)$ ,  $M_r(\beta)$ ,  $M_r(\kappa)$  and  $M_r(\alpha)$  under the some perturbation schemes for the fitted bivariate BSAL model, where  $M_r(\cdot) = C_r(\cdot)/\sum_{i=1}^n C_r(\cdot)$ . From these figures, were detected as potentially influential the following observations: #4, #8, #10, #11, #15, #20, #21, #22, #23, #29 and #35. The observations #8, #11, #15, #20, #22 and #35 correspond to a female patients that has the a pair of a long and a short periods of time until the occurence of failure or censoring. The observation #4 refers to a female patient that has the a pair of the longs periods of time until the occurence of the failure. The observations #23 and #29 have a pair of short periods



**Figure 5.** Index plots of  $M_r(\cdot)$  for: (a)  $\theta$ , (b)  $\beta$ , (c)  $\kappa$  and (d)  $\alpha$ , under joint perturbation in the response variable.



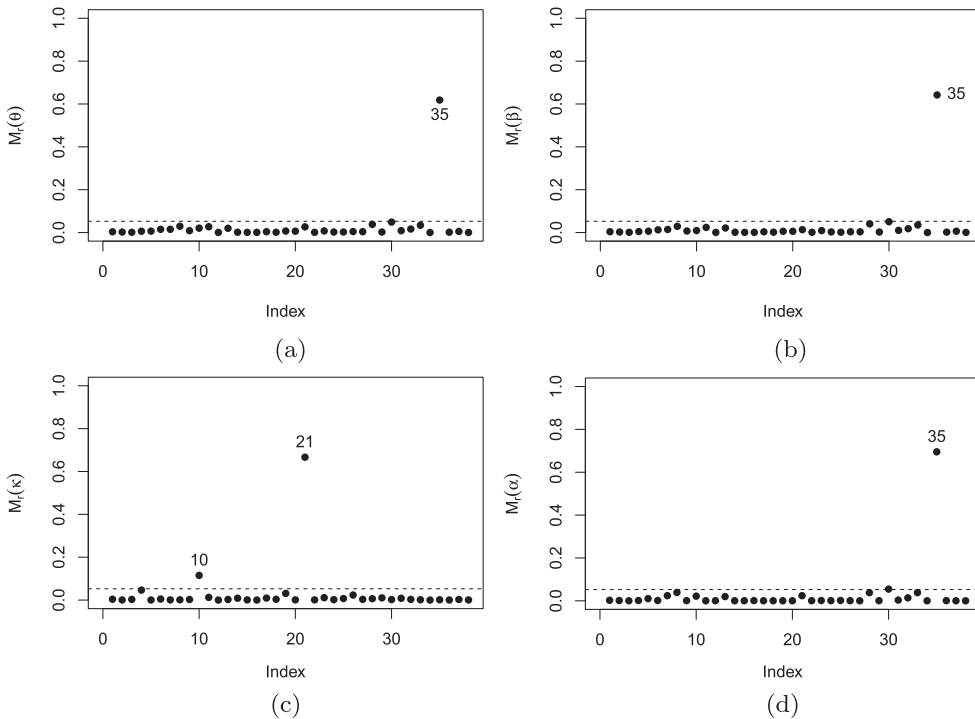


**Figure 6.** Index plots of  $M_r(\cdot)$  for: (a)  $\theta$ , (b)  $\beta$ , (c)  $\kappa$  and (d)  $\alpha$ , under perturbation in the response variable (first coordinate).

of time until failure occurs, where observation #23 refers to a female patient and observation #29 refers to a male patient. Finally, the observations #10 and #21 correspond to a male patients, where observation #10 shows one of the lowest times until the first infection occurs, and observation #21 refers to the longest time until the second infection occurs.

## 6. Discussion

In this paper, we proposed the bivariate BSAL model with dependence structure modeled using frailty. We presented some residuals and obtained the appropriate matrices for assessing the local influence under different perturbation schemes. A simulation study for the residuals was performed to verify if the considered residuals were able to identify misspecifications. We observed that such residuals were not sensitive only to evidence misspecifications in the stochastic component of the model. As showed in the application we compared the proposed model with structure of dependence through the frailty with the model assuming independence within the pairs. We also highlighted the importance of the local influence methodology in detecting points that exercise disproportionate impacts on model parameter estimates and the importance of including dependence on statistical analysis.



**Figure 7.** Index plots of  $M_r(\cdot)$  for: (a)  $\theta$ , (b)  $\beta$ , (c)  $\kappa$  and (d)  $\alpha$ , under perturbation in the response variable (second coordinate).

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## Disclosure statement

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## Appendix

### A.1 Score vector

The expressions used in the definition of the score vector are given by  $\mathbf{D}(\mathbf{a}_j) = \text{diag}\{a_{1j}(\mathbf{w}_1^\top, \boldsymbol{\theta}), \dots, a_{nj}(\mathbf{w}_n^\top, \boldsymbol{\theta})\}$ ,  $\mathbf{D}(\mathbf{b}) = \text{diag}\{b_1(\mathbf{w}_1^\top, \boldsymbol{\theta}), \dots, b_n(\mathbf{w}_n^\top, \boldsymbol{\theta})\}$  and  $\mathbf{D}(\mathbf{c}) = \text{diag}\{c_1(\mathbf{w}_1^\top, \boldsymbol{\theta}), \dots, c_n(\mathbf{w}_n^\top, \boldsymbol{\theta})\}$ , where

$$\begin{aligned} a_{ij}(\mathbf{w}_i^\top, \boldsymbol{\theta}) &= \delta_{i1}\delta_{i2} \frac{1}{f(w_{i1}, w_{i2})} \frac{\partial f(w_{i1}, w_{i2})}{\partial w_{ij}} + \delta_{i1}(1 - \delta_{i2}) \left[ \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i1}}}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}}} \right] \\ &\quad + (1 - \delta_{i1})\delta_{i2} \left[ \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i2}}}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}}} \right] + (1 - \delta_{i1})(1 - \delta_{i2}) \frac{1}{S(w_{i1}, w_{i2})} \\ &\quad \times \frac{\partial S(w_{i1}, w_{i2})}{\partial w_{ij}}, \end{aligned} \quad (\text{A1})$$

$$\begin{aligned} b_i(\mathbf{w}_i^\top, \boldsymbol{\theta}) &= \delta_{i1}\delta_{i2} \frac{1}{f(w_{i1}, w_{i2})} \frac{\partial f(w_{i1}, w_{i2})}{\partial \kappa} + \delta_{i1}(1 - \delta_{i2}) \left[ \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \kappa \partial w_{i1}}}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}}} \right] \\ &\quad + (1 - \delta_{i1})\delta_{i2} \left[ \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \kappa \partial w_{i2}}}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}}} \right] + (1 - \delta_{i1})(1 - \delta_{i2}) \frac{1}{S(w_{i1}, w_{i2})} \\ &\quad \times \frac{\partial S(w_{i1}, w_{i2})}{\partial \kappa}, \end{aligned}$$

and

$$\begin{aligned} c_i(\mathbf{w}_i^\top, \boldsymbol{\theta}) &= \delta_{i1}\delta_{i2} \frac{1}{f(w_{i1}, w_{i2})} \frac{\partial f(w_{i1}, w_{i2})}{\partial \alpha} + \delta_{i1}(1 - \delta_{i2}) \left[ \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \alpha \partial w_{i1}}}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}}} \right] \\ &\quad + (1 - \delta_{i1})\delta_{i2} \left[ \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \alpha \partial w_{i2}}}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}}} \right] + (1 - \delta_{i1})(1 - \delta_{i2}) \frac{1}{S(w_{i1}, w_{i2})} \end{aligned}$$

$$\times \frac{\partial S(w_{i1}, w_{i2})}{\partial \alpha}, \quad (\text{A2})$$

with  $\mathbf{w}_i = (w_{i1}, w_{i2})^\top$ , for  $i = 1, \dots, n$  and  $j = 1, 2$ . The expressions used in Equations (A1) and (A2) are defined below.

$$\begin{aligned} \frac{\partial f(w_{i1}, w_{i2})}{\partial w_{i1}} &= \mathcal{L}''' \{-\log(1 - \Phi(\xi_{21})) - \log[1 - \Phi(\xi_{22})]\} \xi_{12} K(\xi_{22}) \xi_{11}^2 \\ &\quad \times K^2(\xi_{21}) + \mathcal{L}'' \{-\log(1 - \Phi(\xi_{21})) - \log[1 - \Phi(\xi_{22})]\} K(\xi_{22}) \\ &\quad \times \xi_{12} \left[ K^2(\xi_{21}) \xi_{11}^2 + \frac{\xi_{21}(1 - \xi_{11}^2)K(\xi_{21})}{2} \right], \\ \frac{\partial f(w_{i1}, w_{i2})}{\partial w_{i2}} &= \mathcal{L}''' \{-\log(1 - \Phi(\xi_{21})) - \log[1 - \Phi(\xi_{22})]\} \xi_{11} K(\xi_{21}) \xi_{12}^2 \\ &\quad \times K^2(\xi_{22}) + \mathcal{L}'' \{-\log(1 - \Phi(\xi_{21})) - \log[1 - \Phi(\xi_{22})]\} K(\xi_{21}) \\ &\quad \times \xi_{11} \left[ K^2(\xi_{22}) \xi_{12}^2 + \frac{\xi_{22}(1 - \xi_{12}^2)K(\xi_{22})}{2} \right], \\ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij}^2} &= \mathcal{L}'' \{-\log(1 - \Phi(\xi_{21})) - \log[1 - \Phi(\xi_{22})]\} [\xi_{1j} K(\xi_{2j})]^2 \\ &\quad + \mathcal{L}' \{-\log(1 - \Phi(\xi_{21})) - \log[1 - \Phi(\xi_{22})]\} \left[ (\xi_{1j} K(\xi_{2j}))^2 \right. \\ &\quad \left. + \frac{\xi_{2j}(1 - \xi_{1j}^2)K(\xi_{2j})}{2} \right], \\ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i1} \partial w_{i2}} &= \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i2} \partial w_{i1}} = \mathcal{L}'' \{-\log(1 - \Phi(\xi_{21})) - \log[1 - \Phi(\xi_{22})]\} \\ &\quad \times \xi_{11} \xi_{12} K(\xi_{21}) K(\xi_{22}), \\ \frac{\partial f(w_{i1}, w_{i2})}{\partial \alpha} &= -\frac{2\xi_{11}\xi_{12}}{\alpha} \mathcal{L}''' \{-\log(1 - \Phi(\xi_{21})) - \log[1 - \Phi(\xi_{22})]\} [K(\xi_{22}) \\ &\quad \times \xi_{21} K^2(\xi_{21}) + \xi_{22} K(\xi_{21}) K^2(\xi_{22})] + \mathcal{L}'' \{-\log(1 - \Phi(\xi_{21})) \\ &\quad - \log[1 - \Phi(\xi_{22})]\} \{[\xi_{11}(\xi_{21}^2 - 1)K(\xi_{21}) - 2\xi_{11}\xi_{21}K^2(\xi_{21})] \\ &\quad \times \frac{1}{\alpha} \xi_{12} K(\xi_{22}) + \frac{1}{\alpha} \xi_{11} K(\xi_{21}) [\xi_{12}(\xi_{22}^2 - 1)K(\xi_{22}) - 2\xi_{12}\xi_{22} \\ &\quad \times K^2(\xi_{22})]\}, \\ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \alpha \partial w_{ij}} &= -\frac{2}{\alpha} \mathcal{L}'' \{-\log(1 - \Phi(\xi_{21})) - \log[1 - \Phi(\xi_{22})]\} \xi_{1j} K(\xi_{2j}) \\ &\quad \times [(\xi_{21} K(\xi_{21}) + \xi_{22} K(\xi_{22}))] + \frac{1}{\alpha} \mathcal{L}' \{-\log(1 - \Phi(\xi_{21})) \\ &\quad - \log[1 - \Phi(\xi_{22})]\} \left[ \xi_{1j} K(\xi_{2j}) (\xi_{2j}^2 - 1 - 2\xi_{2j} K(\xi_{2j})) \right]. \end{aligned}$$

## A.2 Observed information matrix

The observed information matrix is given by

$$\ddot{\mathbf{L}}_{\theta\theta}^{-1} = - \begin{bmatrix} \ddot{\mathbf{L}}_{\beta\beta} & \ddot{\mathbf{L}}_{\beta\kappa} & \ddot{\mathbf{L}}_{\beta\alpha} \\ \ddot{\mathbf{L}}_{\kappa\beta} & \ddot{\mathbf{L}}_{\kappa\kappa} & \ddot{\mathbf{L}}_{\kappa\alpha} \\ \ddot{\mathbf{L}}_{\alpha\beta} & \ddot{\mathbf{L}}_{\alpha\kappa} & \ddot{\mathbf{L}}_{\alpha\alpha} \end{bmatrix},$$

being

$$\ddot{\mathbf{L}}_{\beta\beta} = \left[ \mathbf{X}_1^\top \mathbf{D}(\mathbf{A}_{11}) + \mathbf{X}_2^\top \mathbf{D}(\mathbf{A}_{12}) \right] \mathbf{X}_1 + \left[ \mathbf{X}_1^\top \mathbf{D}(\mathbf{A}_{21}) + \mathbf{X}_2^\top \mathbf{D}(\mathbf{A}_{22}) \right] \mathbf{X}_2,$$

$$\ddot{\mathbf{L}}_{\beta\kappa} = \ddot{\mathbf{L}}_{\kappa\beta} = - \left[ \mathbf{X}_1^\top \mathbf{D}(\mathbf{d}_1) + \mathbf{X}_2^\top \mathbf{D}(\mathbf{d}_2) \right] \mathbf{Z},$$

$$\ddot{\mathbf{L}}_{\beta\alpha} = \ddot{\mathbf{L}}_{\alpha\beta} = - \left[ \mathbf{X}_1^\top \mathbf{D}(\mathbf{e}_1) + \mathbf{X}_2^\top \mathbf{D}(\mathbf{e}_2) \right] \mathbf{Z},$$

$$\ddot{\mathbf{L}}_{\kappa\kappa} = \text{tr}(\mathbf{D}(\mathbf{f})),$$

$$\ddot{\mathbf{L}}_{\kappa\alpha} = \ddot{\mathbf{L}}_{\alpha\kappa} = \text{tr}(\mathbf{D}(\mathbf{g})),$$

$$\ddot{\mathbf{L}}_{\alpha\alpha} = \text{tr}(\mathbf{D}(\mathbf{h})),$$

where  $\mathbf{D}(\mathbf{A}_{1j}) = \text{diag}\{A_{11}^j, \dots, A_{n1}^j\}$ ,  $\mathbf{D}(\mathbf{A}_{2j}) = \text{diag}\{A_{12}^j, \dots, A_{n2}^j\}$ ,  $\mathbf{D}(\mathbf{d}_j) = \text{diag}\{\partial/\partial\kappa[a_{1j}(\mathbf{w}_1^\top, \boldsymbol{\theta})], \dots, \partial/\partial\kappa[a_{nj}(\mathbf{w}_n^\top, \boldsymbol{\theta})]\}$ ,  $\mathbf{D}(\mathbf{e}_j) = \text{diag}\{\partial/\partial\alpha[a_{1j}(\mathbf{w}_1^\top, \boldsymbol{\theta})], \dots, \partial/\partial\alpha[a_{nj}(\mathbf{w}_n^\top, \boldsymbol{\theta})]\}$ ,  $\mathbf{D}(\mathbf{f}) = \text{diag}\{\partial/\partial\kappa[b_1(\mathbf{w}_1^\top, \boldsymbol{\theta})], \dots, \partial/\partial\kappa[b_n(\mathbf{w}_n^\top, \boldsymbol{\theta})]\}$ ,  $\mathbf{D}(\mathbf{g}) = \text{diag}\{\partial/\partial\alpha[c_1(\mathbf{w}_1^\top, \boldsymbol{\theta})], \dots, \partial/\partial\alpha[c_n(\mathbf{w}_n^\top, \boldsymbol{\theta})]\}$  and  $\mathbf{D}(\mathbf{h}) = \text{diag}\{\partial/\partial\alpha[c_1(\mathbf{w}_1^\top, \boldsymbol{\theta})], \dots, \partial/\partial\alpha[c_n(\mathbf{w}_n^\top, \boldsymbol{\theta})]\}$ , with

$$\begin{aligned} A_{ij}^j &= \frac{\partial a_{ij}(\mathbf{w}_i^\top, \boldsymbol{\theta})}{\partial w_{ij}} = \delta_{i1}\delta_{i2} \left[ - \left( \frac{\frac{\partial f(w_{i1}, w_{i2})}{\partial w_{ij}}}{f(w_{i1}, w_{i2})} \right)^2 + \frac{1}{f(w_{i1}, w_{i2})} \frac{\partial^2 f(w_{i1}, w_{i2})}{\partial w_{ij}^2} \right] \\ &\quad + \delta_{i1}(1 - \delta_{i2}) \left[ \frac{1}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}}} \frac{\partial}{\partial w_{ij}} \left( \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i1}} \right) \right. \\ &\quad \left. - \left( \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i1}}}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}}} \right)^2 \right] + \delta_{i2}(1 - \delta_{i1}) \left[ - \left( \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i2}}}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}}} \right)^2 \right. \\ &\quad \left. + \frac{1}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}}} \frac{\partial}{\partial w_{ij}} \left( \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i2}} \right) \right] + (1 - \delta_{i1})(1 - \delta_{i2}) \\ &\quad \times \left[ - \left( \frac{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{ij}}}{S(w_{i1}, w_{i2})} \right)^2 + \frac{1}{S(w_{i1}, w_{i2})} \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij}^2} \right] \end{aligned}$$

and

$$\begin{aligned}
 A_{ij}^{j'} = \frac{\partial a_{ij}(\mathbf{w}_i^\top, \boldsymbol{\theta})}{\partial w_{ij'}} = & \delta_{i1} \left[ -\frac{\frac{\partial f(w_{i1}, w_{i2})}{\partial w_{ij'}} \frac{\partial f(w_{i1}, w_{i2})}{\partial w_{ij}}}{f^2(w_{i1}, w_{i2})} + \frac{1}{f(w_{i1}, w_{i2})} \frac{\partial^2 f(w_{i1}, w_{i2})}{\partial w_{ij'} \partial w_{ij}} \right] \\
 & \times \delta_{i2} + (1 - \delta_{i2}) \delta_{i1} \left[ -\frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij'} \partial w_{i1}} \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i1}}}{\left( \frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}} \right)^2} + \frac{1}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}}} \right. \\
 & \times \frac{\partial}{\partial w_{ij'}} \left( \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i1}} \right) \left. + \delta_{i2} (1 - \delta_{i1}) \left[ \frac{1}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}}} \right. \right. \\
 & \times \frac{\partial}{\partial w_{ij'}} \left( \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i2}} \right) - \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij'} \partial w_{i2}} \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i2}}}{\left( \frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}} \right)^2} \left. \right] \\
 & + (1 - \delta_{i1})(1 - \delta_{i2}) \left[ -\frac{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{ij'}} \frac{\partial S(w_{i1}, w_{i2})}{\partial w_{ij}}}{S^2(w_{i1}, w_{i2})} + \frac{1}{S(w_{i1}, w_{i2})} \right. \\
 & \times \left. \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij'} \partial w_{ij}} \right],
 \end{aligned}$$

where  $j \neq j'$ . Furthermore,

$$\begin{aligned}
 \frac{\partial a_{ij}(\mathbf{w}_i^\top, \boldsymbol{\theta})}{\partial \alpha} = & \delta_{i1} \delta_{i2} \left[ -\frac{\frac{\partial f(w_{i1}, w_{i2})}{\partial \alpha} \frac{\partial f(w_{i1}, w_{i2})}{\partial w_{ij}}}{f^2(w_{i1}, w_{i2})} + \frac{1}{f(w_{i1}, w_{i2})} \frac{\partial^2 f(w_{i1}, w_{i2})}{\partial \alpha \partial w_{ij}} \right] \\
 & + \delta_{i1} (1 - \delta_{i2}) \left[ -\frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \alpha \partial w_{i1}} \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i1}}}{\left( \frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}} \right)^2} + \frac{\partial}{\partial \alpha} \left( \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i1}} \right) \right. \\
 & \times \left. \frac{1}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}}} \right] + (1 - \delta_{i1}) \delta_{i2} \left[ \frac{1}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}}} \frac{\partial}{\partial \alpha} \left( \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i2}} \right) \right. \\
 & \times \left. \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \alpha \partial w_{i2}} \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij} \partial w_{i2}}}{\left( \frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}} \right)^2} \right] + (1 - \delta_{i1})(1 - \delta_{i2}) \left[ \frac{1}{S(w_{i1}, w_{i2})} \right.
 \end{aligned}$$

$$\times \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \alpha \partial w_{ij}} - \frac{\frac{\partial S(w_{i1}, w_{i2})}{\partial \alpha} \frac{\partial S(w_{i1}, w_{i2})}{\partial w_{ij}}}{S^2(w_{i1}, w_{i2})} \right]$$

and

$$\begin{aligned} \frac{\partial c_i(\mathbf{w}_i^\top, \boldsymbol{\theta})}{\partial \alpha} = & \delta_{i1} \delta_{i2} \left[ - \left( \frac{\frac{\partial f(w_{i1}, w_{i2})}{\partial \alpha}}{f(w_{i1}, w_{i2})} \right)^2 + \frac{1}{f(w_{i1}, w_{i2})} \frac{\partial^2 f(w_{i1}, w_{i2})}{\partial \alpha^2} \right] \\ & \times (1 - \delta_{i2}) \left[ - \left( \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \alpha \partial w_{i1}}}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}}} \right)^2 + \frac{\partial}{\partial \alpha} \left( \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \alpha \partial w_{i1}} \right) \right. \\ & \times \left. \frac{1}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i1}}} \right] + (1 - \delta_{i1}) \delta_{i2} \left[ \frac{1}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}}} \frac{\partial}{\partial \alpha} \left( \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \alpha \partial w_{i2}} \right) \right. \\ & \times \left. - \left( \frac{\frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \alpha \partial w_{i2}}}{\frac{\partial S(w_{i1}, w_{i2})}{\partial w_{i2}}} \right)^2 \right] + (1 - \delta_{i1})(1 - \delta_{i2}) \left[ - \left( \frac{\frac{\partial S(w_{i1}, w_{i2})}{\partial \alpha}}{S(w_{i1}, w_{i2})} \right)^2 \right. \\ & \times \left. + \frac{1}{S(w_{i1}, w_{i2})} \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial \alpha^2} \right], \end{aligned}$$

with

$$\begin{aligned} \frac{\partial^2 f(w_{i1}, w_{i2})}{\partial w_{ij}^2} = & \xi_{i1} \xi_{i2} K(\xi_{21}) K(\xi_{22}) \left\{ \mathcal{L}^{\text{iv}} \{ -\log [1 - \Phi(\xi_{21})] - \log [1 - \Phi(\xi_{22})] \} \right. \\ & \times \left[ \xi_{1j} K(\xi_{2j}) \right]^2 + \mathcal{L}^{\text{iv}} \{ -\log [1 - \Phi(\xi_{21})] - \log [1 - \Phi(\xi_{22})] \} \left[ \frac{1}{2} \xi_{2j} \right. \\ & \times \left. \left( K(\xi_{2j}) - \frac{1}{2} \xi_{2j} \right) \right] 2K(\xi_{2j}) + \left[ \xi_{1j} K(\xi_{2j}) \right]^2 + \frac{1}{2} \xi_{2j} (1 - \xi_{1j}^2) \\ & \times K(\xi_{2j}) \left. \right\} + \mathcal{L}^{\text{iv}} \{ -\log [1 - \Phi(\xi_{21})] - \log [1 - \Phi(\xi_{22})] \} \left\{ 2K(\xi_{2j}) \right. \\ & \times \left[ \frac{1}{2} \xi_{2j} + \xi_{1j}^2 \left( K(\xi_{2j}) - \frac{1}{2} \xi_{2j} \right) \right] + \frac{1}{4} \left[ 1 - \xi_{1j}^2 - 2\xi_{2j}^2 + (1 - \xi_{1j}^2) \right. \\ & \times \left. \left. 2\xi_{2j} \left( K(\xi_{2j}) - \frac{1}{2} \xi_{2j} \right) \right] \right\}, \end{aligned}$$

$$\begin{aligned} \frac{\partial}{\partial w_{i2}} \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i2} \partial w_{i1}} \right] = & \xi_{i1} K(\xi_{21}) \left\{ \mathcal{L}^{\text{iv}} \{ -\log [1 - \Phi(\xi_{21})] - \log [1 - \Phi(\xi_{22})] \} \right. \\ & \times \left[ \xi_{i2} K(\xi_{22}) \right]^2 + \mathcal{L}^{\text{iv}} \{ -\log [1 - \Phi(\xi_{21})] - \log [1 - \Phi(\xi_{22})] \} \\ & \times K(\xi_{22}) \left[ \frac{1}{2} \xi_{22} + \xi_{i2}^2 \left( K(\xi_{22}) - \frac{1}{2} \xi_{22} \right) \right] \left. \right\}, \end{aligned}$$



$$\begin{aligned}
\frac{\partial}{\partial w_{i1}} \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i1} \partial w_{i2}} \right] &= \xi_{12} K(\xi_{22}) \left\{ \mathcal{L}''' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} \right. \\
&\quad \times [\xi_{11} K(\xi_{21})]^2 + \left[ \frac{1}{2} \xi_{21} + \xi_{11}^2 \left( K(\xi_{21}) - \frac{1}{2} \xi_{21} \right) \right] \\
&\quad \times K(\xi_{21}) \mathcal{L}'' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} \Big\}, \\
\frac{\partial^3 S(w_{i1}, w_{i2})}{\partial w_{ij}^3} &= \xi_{1j} K(\xi_{2j}) \left\{ \mathcal{L}''' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} \xi_{1j}^2 K^2(\xi_{2j}) \right. \\
&\quad + \left[ \frac{3}{2} \xi_{2j} K(\xi_{2j}) (1 - \xi_{1j}^2) + 3 (\xi_{1j} K(\xi_{2j}))^2 \right] \mathcal{L}'' \{-\log[1 - \Phi(\xi_{21})] \\
&\quad - \log[1 - \Phi(\xi_{22})]\} + \mathcal{L}' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} \\
&\quad \times \left\{ \xi_{2j} K(\xi_{2j}) + \left( K(\xi_{2j}) - \frac{1}{2} \xi_{2j} \right) \left[ 2 \xi_{1j}^2 K(\xi_{2j}) + \frac{1}{2} \xi_{2j} (1 - \xi_{1j}^2) \right] \right. \\
&\quad \left. \left. + \frac{1}{4} (1 - \xi_{1j}^2 - 2 \xi_{2j}^2) \right\} \right\},
\end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 f(w_{i1}, w_{i2})}{\partial \alpha \partial w_{ij}} &= \frac{\xi_{1j'} K(\xi_{2j'})}{\alpha} \left\{ \mathcal{L}^{iv} \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} \xi_{1j}^2 \right. \\
&\quad \times K^2(\xi_{2j}) [-2 (\xi_{21} K(\xi_{21}) + \xi_{22} K(\xi_{22}))] + \mathcal{L}''' \{-\log[1 - \Phi(\xi_{21})] \\
&\quad - \log[1 - \Phi(\xi_{22})]\} (\xi_{1j} K(\xi_{2j}))^2 [(\xi_{2j'} (\xi_{2j'} - 2K(\xi_{2j'})) - 1) + 2 \\
&\quad \times (\xi_{2j} (\xi_{2j} - 2K(\xi_{2j})) - 1)] - 2 [\xi_{2j'} K(\xi_{2j'}) + \xi_{2j} K(\xi_{2j})] \\
&\quad \times \mathcal{L}''' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} \left[ (\xi_{1j} K(\xi_{2j}))^2 + \frac{1}{2} \right. \\
&\quad \times \xi_{2j} (1 - \xi_{1j}^2) K(\xi_{2j}) \Big] + \mathcal{L}'' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} \\
&\quad \left\{ [\xi_{2j'} (\xi_{2j'} - 2K(\xi_{2j'})) - 1] \left[ (\xi_{1j} K(\xi_{2j}))^2 + \frac{1}{2} \xi_{2j} (1 - \xi_{1j}^2) K(\xi_{2j}) \right] \right. \\
&\quad + 2 [\xi_{1j} K(\xi_{2j})]^2 [\xi_{2j} (\xi_{2j} - K(\xi_{2j})) - 1] + \frac{1}{2} [\xi_{2j} (3 \xi_{1j}^2 - 2) K(\xi_{2j}) \\
&\quad \left. \left. + \xi_{2j}^2 (1 - \xi_{1j}^2) K(\xi_{2j}) (\xi_{2j} - 2K(\xi_{2j})) \right] \right\},
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial \alpha} \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{ij}^2} \right] &= -\frac{2}{\alpha} (\xi_{1j} K(\xi_{2j}))^2 \left\{ \mathcal{L}''' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} \right. \\
&\quad \times [\xi_{21} K(\xi_{21}) + \xi_{22} K(\xi_{22})] - [\xi_{2j} (\xi_{2j} - 2K(\xi_{2j})) - 1] \\
&\quad \times \mathcal{L}'' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} \Big\} - \frac{2}{\alpha} \\
&\quad \times \mathcal{L}'' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} (\xi_{21} K(\xi_{21}) \\
&\quad + \xi_{22} K(\xi_{22})) \left[ (\xi_{1j} K(\xi_{2j}))^2 + \frac{1}{2} \xi_{2j} (1 - \xi_{1j}^2) K(\xi_{2j}) \right] + \frac{1}{\alpha} \\
&\quad \times \mathcal{L}' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} \{ 2 K^2(\xi_{2j}) \}
\end{aligned}$$

$$\times \xi_{1j}^2 [\xi_{2j}(\xi_{2j} - 2K(\xi_{2j})) - 1] + \frac{1}{2} \left[ K(\xi_{2j})\xi_{2j} \left( (3\xi_{1j}^2 - 1) \right. \right. \\ \left. \left. + \xi_{2j}(1 - \xi_{1j}^2)(\xi_{2j} - 2K(\xi_{2j})) \right) \right] \Bigg\},$$

$$\frac{\partial}{\partial \alpha} \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i1} \partial w_{i2}} \right] = \frac{\partial}{\partial \alpha} \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i2} \partial w_{i1}} \right] = \frac{1}{\alpha} \xi_{11} \xi_{12} K(\xi_{21}) K(\xi_{22}) \{-2(\xi_{21} K(\xi_{21}) \\ + \xi_{22} K(\xi_{22})) \mathcal{L}''' \{-\log[1 - \Phi(\xi_{21})] \\ - \log[1 - \Phi(\xi_{22})]\} + [\xi_{21}(\xi_{21} - 2K(\xi_{21})) \\ + \xi_{22}(\xi_{22} - 2K(\xi_{22})) - 2]\},$$

$$\frac{\partial^2 f(w_{i1}, w_{i2})}{\partial \alpha^2} = \frac{1}{\alpha^2} \xi_{11} \xi_{12} K(\xi_{21}) K(\xi_{22}) \{[\xi_{21}(\xi_{21} - 2K(\xi_{21})) + \xi_{22}(\xi_{22} - 2K(\xi_{22})) - 2] \\ \times [-2\mathcal{L}''' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} (\xi_{21} K(\xi_{21}) + \xi_{22} K(\xi_{22})) \\ + \mathcal{L}'' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} [\xi_{21}(\xi_{21} - 2K(\xi_{21})) + \xi_{22}(\xi_{22} \\ - 2K(\xi_{22})) - 2] + 4(\xi_{21} K(\xi_{21}) + \xi_{22} K(\xi_{22}))^2 \mathcal{L}^{\text{iv}} \{-\log[1 - \Phi(\xi_{21})] \\ - \log[1 - \Phi(\xi_{22})]\} - 2\mathcal{L}''' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} [K(\xi_{21}) \\ \times \xi_{21} [\xi_{21}(\xi_{21} - 2K(\xi_{21})) - 1] + \xi_{22} K(\xi_{22}) [\xi_{22}(\xi_{22} - 2K(\xi_{22})) - 1]] \\ - 2\mathcal{L}''' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} (\xi_{21} K(\xi_{21}) + \xi_{22} K(\xi_{22})) \\ \times [\xi_{21}(\xi_{21} - 2K(\xi_{21})) + \xi_{22}(\xi_{22} - 2K(\xi_{22})) - 2] + \mathcal{L}'' \{-\log[1 - \Phi(\xi_{21})] \\ - \log[1 - \Phi(\xi_{22})]\} [-2\xi_{21}(K(\xi_{21})(\xi_{21} - 2K(\xi_{21}) - 1) + \xi_{21}) - 2\xi_{22} \\ \times (K(\xi_{22})(\xi_{22} - 2K(\xi_{22}) - 1) + \xi_{22})]\},$$

$$\frac{\partial^3 S(w_{i1}, w_{i2})}{\partial \alpha^2 \partial w_{ij}} = \frac{1}{\alpha^2} \xi_{1j} K(\xi_{2j}) \{[\xi_{2j}(\xi_{2j} - 2K(\xi_{2j})) - 2] [-2(\xi_{21} K(\xi_{21}) + \xi_{22} K(\xi_{22})) \\ \times \mathcal{L}'' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} + (\xi_{2j}^2 - 1 - 2\xi_{2j} K(\xi_{2j})) \\ \times \mathcal{L}' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} + 4\mathcal{L}''' \{-\log[1 - \Phi(\xi_{21})] \\ - \log[1 - \Phi(\xi_{22})]\} (\xi_{21} K(\xi_{21}) + \xi_{22} K(\xi_{22}))^2 - 2\mathcal{L}'' \{-\log[1 - \Phi(\xi_{21})] \\ - \log[1 - \Phi(\xi_{22})]\} [\xi_{21} K(\xi_{21})(\xi_{21}(\xi_{21} - 2K(\xi_{21})) - 1) + \xi_{22} K(\xi_{22})(\xi_{22} \\ \times (\xi_{22} - 2K(\xi_{22})) - 1)] - 2[(\xi_{21} K(\xi_{21}) + \xi_{22} K(\xi_{22})) (\xi_{2j}^2 - 2K(\xi_{2j}) \\ \times \xi_{2j} - 1) \mathcal{L}'' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\} + (\xi_{2j}^2 + K(\xi_{2j})\xi_{2j} \\ \times (\xi_{2j}(\xi_{2j} - 2K(\xi_{2j})) - 1)) \mathcal{L}' \{-\log[1 - \Phi(\xi_{21})] - \log[1 - \Phi(\xi_{22})]\}]\}.$$

In addition, we have

$$\frac{\partial}{\partial w_{i2}} \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i2} \partial w_{i1}} \right] = \frac{\partial}{\partial w_{i2}} \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i1} \partial w_{i2}} \right] = \frac{\partial}{\partial w_{i1}} \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i2}^2} \right]$$

and

$$\frac{\partial}{\partial w_{i1}} \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i1} \partial w_{i2}} \right] = \frac{\partial}{\partial w_{i1}} \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i2} \partial w_{i1}} \right] = \frac{\partial}{\partial w_{i2}} \left[ \frac{\partial^2 S(w_{i1}, w_{i2})}{\partial w_{i1}^2} \right].$$